



Lifting Tool for Single Grouser Shoes Install & Removal on D6, D9, D10 & D11 Series TTT

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Lifting Tool for Single Grouser Shoes Install & Removal on D6, D9, D10 & D11 Series TTT.....	0
1.0 Introduction.....	1
2.0 Best Practice Description.....	1
3.0 Implementation Steps.....	3
4.0 Benefits.....	6
5.0 Resources Required.....	6
6.0 Supporting Attachments / References.....	6
7.0 Related Best Practices.....	9
8.0 Acknowledgements.....	9

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1.0 Introduction

Due to their irregular geometry, weight, and quantity by equipment, the track shoes are one of the most difficult and awkward components to handle during the track assembly and disassembly operations on Track Type Tractors. The track shoe weights vary according to the model, ranging from 30kg (D6 series) to 85kg (D11 series) in large groups from 41 to 53 track shoes in the D11 series and D9 series respectively. This results in a slow, risky, and stressful process for the technical personnel involved in the handling of this component.

This Best Practice provides the design of a tool that allows easy manipulation of the track shoes in any process that involves lifting, turning and moving of this.

Editor's Note: The use of this installation tool and the processes discussed within this Best Practice is not a substitute for applicable laws, standards and regulations and does not alter or limit the obligation of individuals who utilize the tool to fully comply with federal, state and local law and prudent safety measures relating to the use of lifting devices.

2.0 Best Practice Description

For the lifting and handling of the track shoes in any install and removal operations; a lift chain assembly is currently used (see Fig. 1).

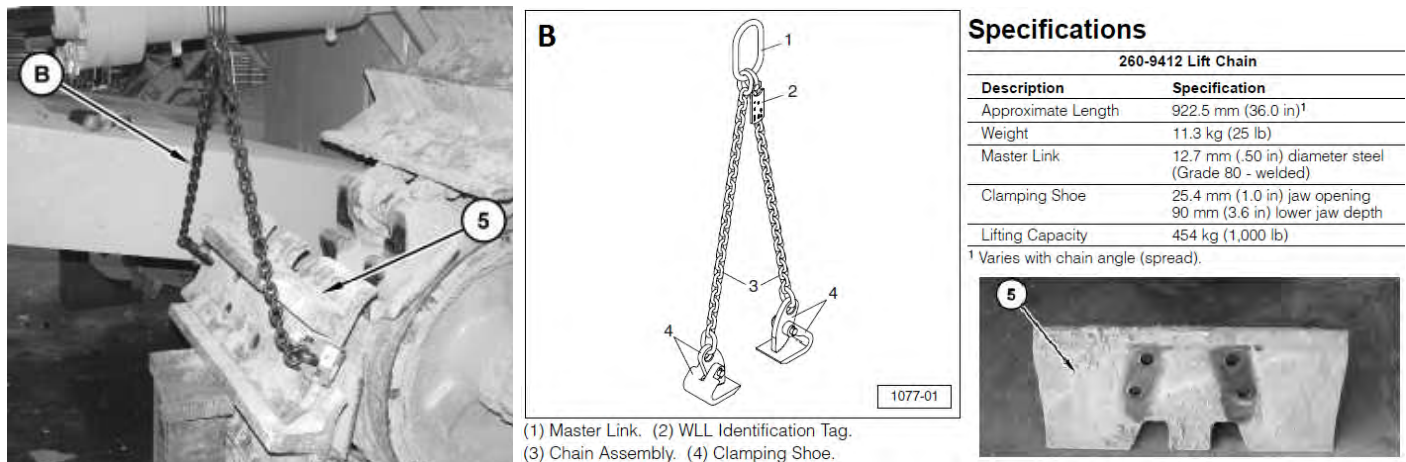


Fig.1. Lifting equipment previously used for track shoes.

Some lifting equipment similar to this one have certain safety disadvantages during the operation when turning the track shoe to an angle higher than 15 ° and up to 30 ° on the imaginary axis which closes the lifting triangle (see left photograph, Fig. 1.) Because this Clamping Shoe (see component # 4 Fig. 1) is a horizontal lifting device for materials having thin and flat gripping areas; these considerably lose their ability to grasp the angles exceed aforementioned (see image B, Figure 2.).

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Lifting Tool for single grouser shoes install and removal on D6; D9; D10 & D11 series TTT	DATE 4/20/2016	CHG NO 01	NUMBER 1401-3.04-1331



Fig.2. Maximum allowed angle for horizontal clamps¹

On the other hand, the use of this device demands some effort from the technician in order to bend the tracks and overcome the horizontal resting position of the hanging shoe, when seeking the alignment position of the shoe and the master link holes as well for the final installation of the bolts.

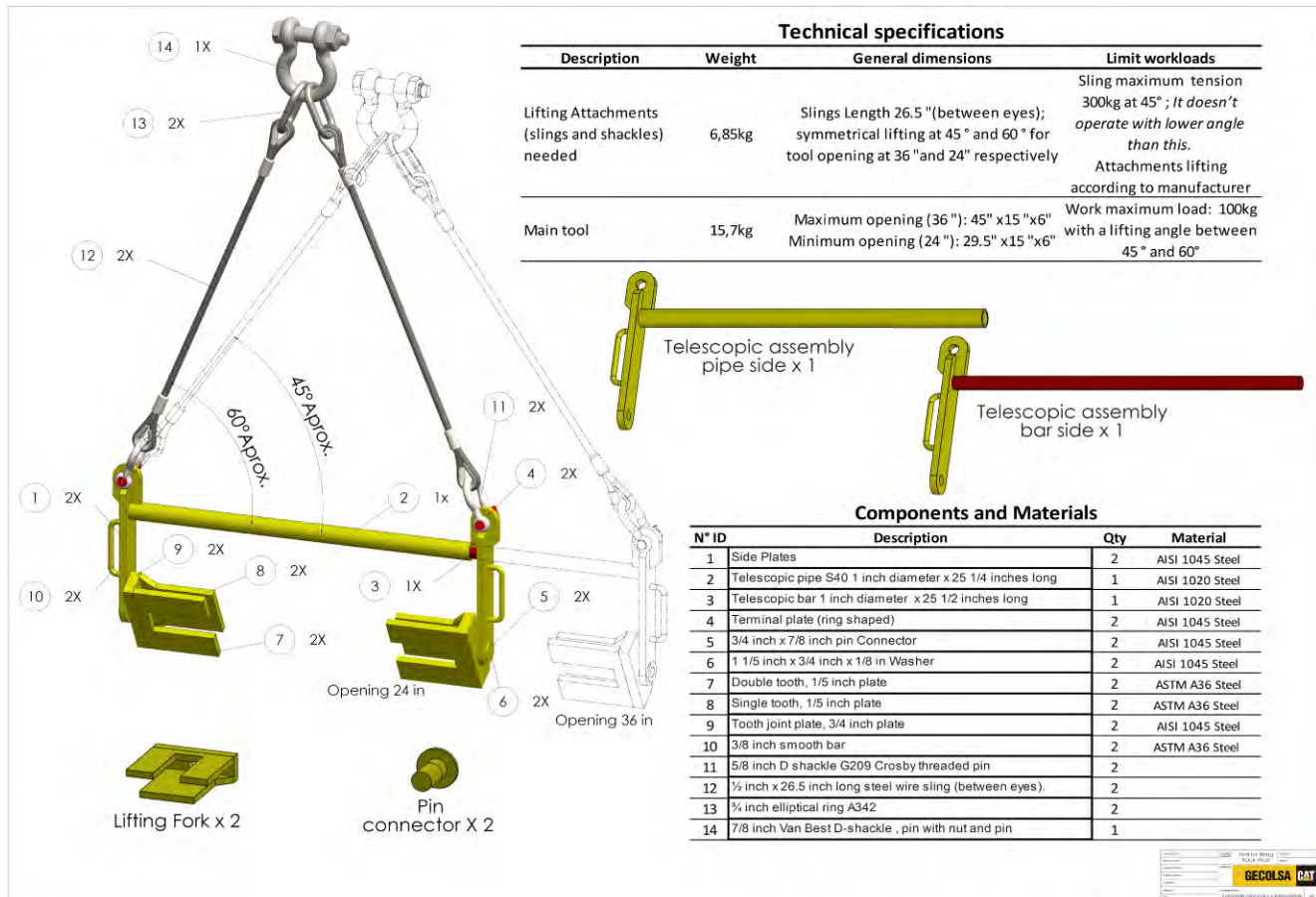


Fig.3. New proposed track shoe lifting tool

¹ CrosbyIP Garras para izaje/ http://www.iph.com.ar/descargas/Mordazas_pchapa.pdf. By Crosby Group, Inc. 2006

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DATE
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CHG
NO
01

NUMBER
1401-3.04-1331

The new proposed tool is designed to prevent the drawbacks described previously. It consists of two symmetrical assemblies reinforced with AISI 1045 steel, three teeth which form a clamping grip in ASTM A36 steel connected by a AISI 1045 steel plate. These two assemblies are joined by an ASTM – 1020 steel bar-tube telescopic system, which allows an opening and closing range of the clamps from 24 to 36 inches maximum (measured from the bottom of each clamp) covering all the models of track shoes that are within a length of this range. The three teeth of the clamp secure the track shoe on the lateral profile of the main grouser, allowing easy 180° rotation on the axis of the grouser by means of a ¾ inch pin (AISI 1045) which connects the clamp to the lateral assemblies.

At the upper ends of the tool we can see terminals made of 1045 steel sheet having ¾" diameter bore, as anchor points. These are located to permit the installation and precise auto center of the bolt shackle threaded up to 5/8". This way, the closing pin of these can easily turn over the bore, always having symmetry between the two anchor points. Forming 45° to 60° angles for the opening of a 36" and 24" tool respectively. Two steel wire slings of 26.5" long with a low resistance of 300kg each, close the lifting triangle by means of a master link or shackle of 7/8" (see Figure 3).

On the lateral plates, the tool consists of rectangular rings on a 3/8" rod for the optional installation of ropes as lifting guides and it also serves as clamping handles for manipulating the tool.

3.0 Implementation Steps

- Fabricate the tool according to the drawings, taking into account the selection of the materials given. (See attached drawings).
- Select the lifting accessories (steel slings, shackles, etc.) as recommended by loading regulations listed below, taking into account the availability of local suppliers:
 - o Total Lifting load 120kg
 - o Correct length of sling to keep lifting angle less than 90° for the largest shoe
 - o Symmetrical Lifting
 - o 26.5 " long steel slings
 - o 300kg tension maximum steel slings (each)
 - o Screwed pin lower shackles (See Fig. 3)
 - o Pin upper shackle, nut and safety pin (See Fig. 3). It can also use a master link assembly
- Before attaching the track shoe, check the condition of the tool for cracks, misuse or any defect that do not guarantee its design properties and proper operation. Also check the label or specification plate, making sure the loading capacity covers the weight of the shoe to be hoisted.

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Fig. 4. Holding the shoe and critical points for prior review before using the tool.

- Once you have the lifting equipment complete, including the main tool as shown in Fig. 3. You must extend the telescopic bar enough to exceed the width of the shoe that will be hoisted (See Fig. 4).
- The telescopic bar closes, holding the shoe on the main grouser, so that the lateral profiles of this make contact with the bottom of the grips. For the master link disassembly process, this must be done before completely losing the bolts. It is important to check for safety that the telescopic bar is not up to 12 3/4" out of the pipe (See Fig. 5) during the lifting operation. This distance is identified and marked on the bar.

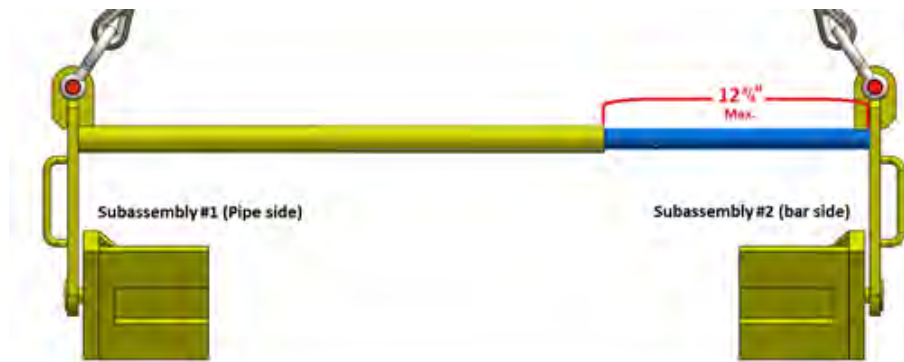


Fig.5. Maximum opening distance of the pipe with lifting load applied.

- The lifting is done only until the shoe hangs freely, at which point the fastening should be double checked and the lifting angles are less than 90°. Also ensure the symmetry, the closing, and automatic lock of the telescopic bar are ok. Finally, proceed to completely lift the shoe and move it where required (See Fig. 6).

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Fig.6. Procedure for lifting and handling of the shoe in the installation and/or removal process.

- To release the shoe, you must ensure that there is no tension in the lifting slings and they must allow to reopen the telescopic bar enough to disengage the shoe. Also, it should already be on a safe surface (disassembly process), or fastened with bolts to the master link (installation process).

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4.0 Benefits

- Practical design easy to manufacture.
- Economically feasible. Tool cost USD \$500 plus lifting attachments costs USD \$250 for a total cost of USD \$750.
- Light tool and easy to transport, as it is easily detachable. The heaviest part is the sub-assembly, which is only 8kg (including clamp, see Fig. 4).
- Clamping claws that allow the shoe to rotate 80 ° (both ways) is on the axis of the grouser. This allows you to easily align the shoe with the master link bores in the installation procedure of the bolts.
- Auto-lock of the telescopic system by symmetric lifting which causes force components into compression between the pipe and the bar ensuring the safe fastening of the shoe.
- Its telescopic design allows to cover shoe models from 24" to 36" inches wide on single grouser shoe.
- It requires less physical effort from the technical personnel for its handling in the disassembly and assembly shoe process, making them safer, comfortable and faster.
- It allows the installation of guiding ropes for a greater control of the movements when lifting, and at the same time these rectangular rings serve as handles for manipulating the sub - assemblies of the tool.

5.0 Resources Required

- Materials and components required (See Fig. 3)
- Design drawings (See Fig. 7A; 7B; 7C)
- Total cost of the tool USD \$750 (including all lifting attachments)
- Lifting equipment (jib crane, bridge crane, etc.) with a minimum capacity of 200kg
- A technician.

6.0 Supporting Attachments / References

- Maximum track shoe weight 85kg approx. (D11 series).

CRITICAL SCENARIO, 36" TOOL OPENING @ MAXIMUM TEST LOAD

ID. ANALISYS	DESCRIPTION	TEST LOADS (kg)	DESIGN FACTOR
A	Sub-assembly. Lifting Fork and connector pin.	135/2»70 + gravity	5,9
B	Sub-assembly. Terminal, side plate and telescopi pipe.	135/2»70 + gravity	35
C	Sub-assembly, Terminal, side plate and Telescopic bar.	135/2»70 + gravity	36
D	Complete assembly of the tool	90*1.5=135 + gravity	7,2

Table 1. Tests loads description and Safety factor results.

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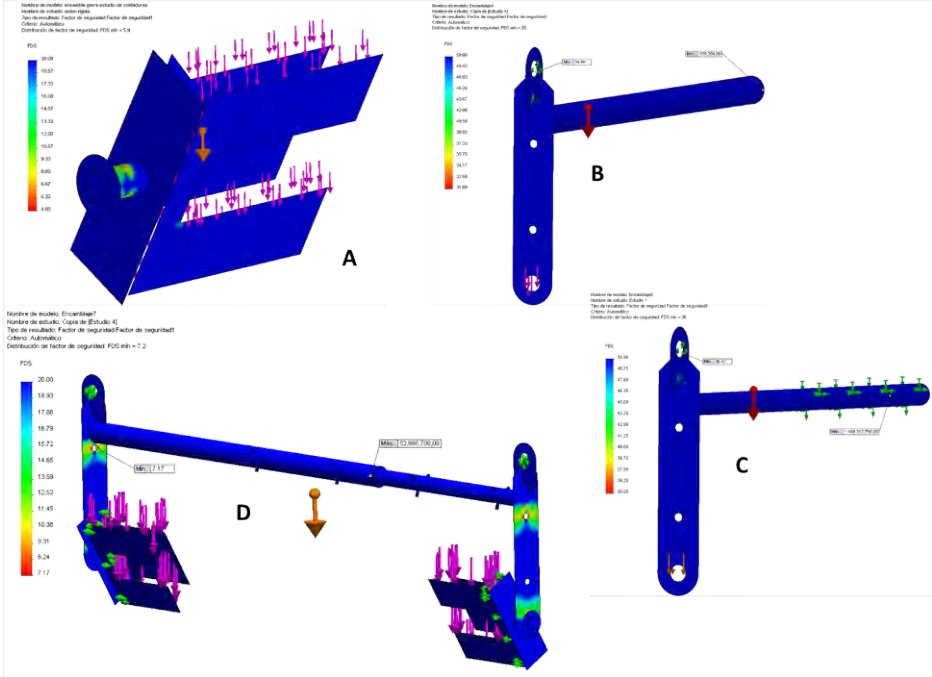


Fig. 6. Design Factor Validation design factor under the ASME B30.20-2003 standard: *Below-the-Hook lifting device*. Minimum 5:1

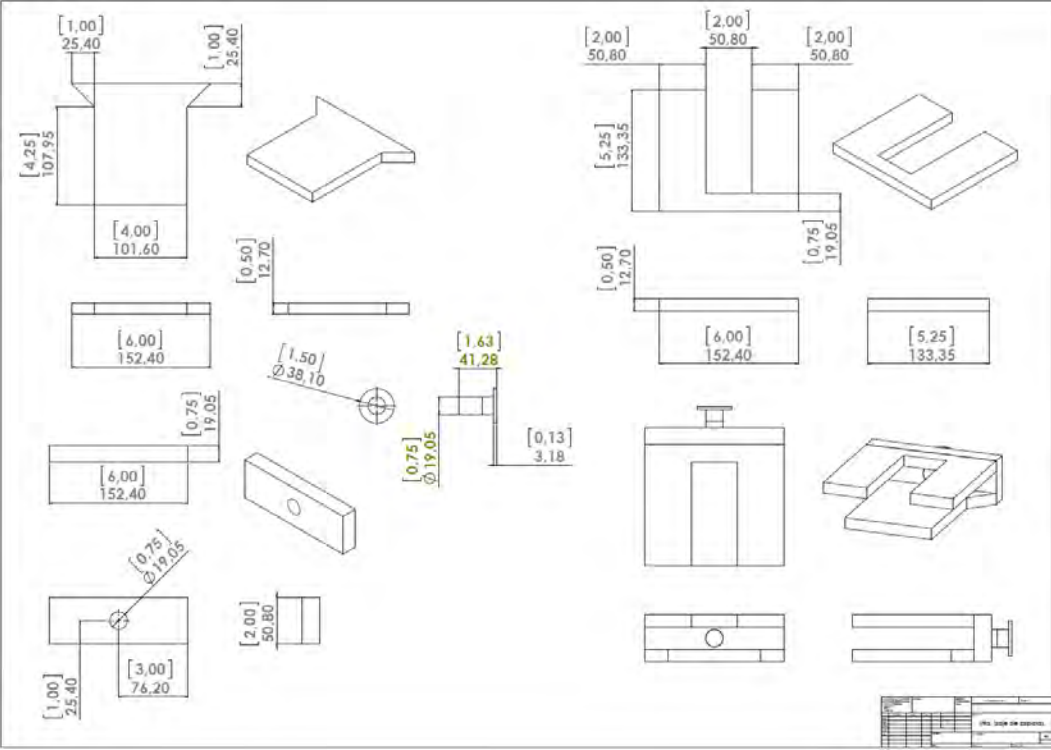


Fig. 7A Tool Design drawings

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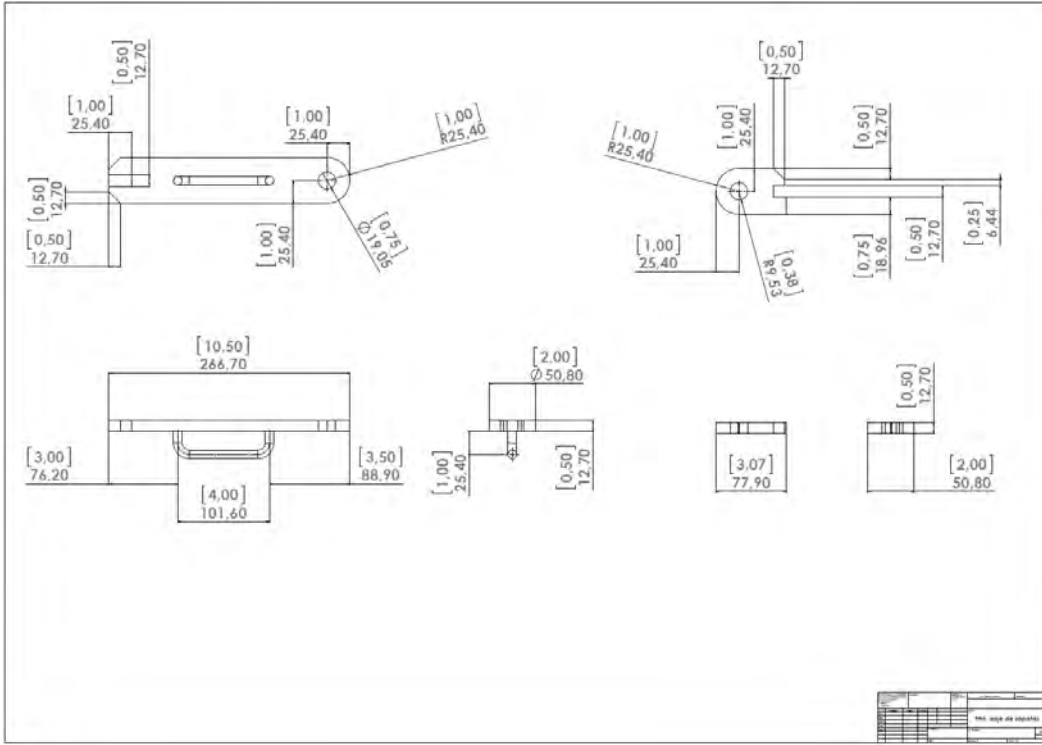


Fig. 7B Tool Design drawings

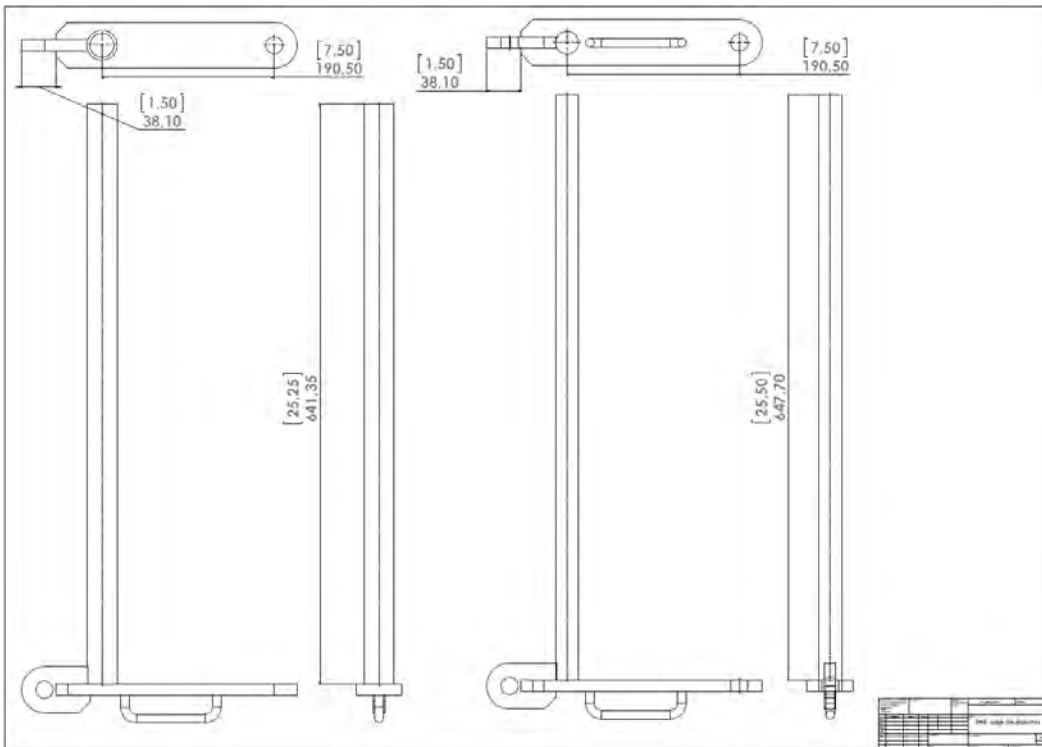


Fig. 7C Tool Design drawings

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7.0 Related Best Practices

None

8.0 Acknowledgements

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